

Massively parallel, single-molecule assessment of synthetic fidelity and drug-like properties in a DNA-encoded library

Authors: Grant Koch¹, Meghan F. Lawler², Adam Murray¹, Neil Carlson², Alexander Engstrom¹, Panpan Zhang¹, Mariel Steiner¹, Sarah Beth Avila¹, Kaileigh Cloutier-Leblanc², Sumudu Leelananda², LaShadric Grady², Jaru Taechalertpaisarn¹, Conor Corcoran¹, Nicolas Tilmans², R. Scott Lokey^{1*}

Affiliations:

*Corresponding author. Email: slokey@ucsc.edu

¹Present address: Department of Chemistry and Biochemistry, University of California, Santa Cruz; 1156 High St., Santa Cruz, CA, 95064

²Present address: Anagenex; 20 Maguire Rd Suite 302, Lexington, MA, 02421

Abstract: DNA-encoded libraries (DELs) have emerged as a promising drug discovery strategy, but successful translation of hits is often impeded by synthetic inefficiency and enrichment of poorly permeable compounds. Here we introduce a novel sequencing-based separation strategy, LC-seq, that simultaneously evaluates synthetic fidelity and permeability-relevant lipophilicity for individual DNA-encoded library members. Using a 120,000-member peptide library, we mapped reaction efficiency across all synthetic cycles and identified structure-reactivity trends. The on-DNA lipophilicities for resynthesized library members correlate strongly with their off-DNA lipophilicities and passive permeability in artificial membranes and MDCK cells. This approach enables direct assessment of compound quality and drug-like properties at unprecedented scale, potentially transforming DEL-based drug discovery.

Introduction. Encoded library technologies accelerate the discovery of protein-targeting molecules by leveraging the vast chemical diversity achieved through the combinatorial permutation of chemical building blocks(1, 2). While display technologies such as mRNA(3, 4) and phage display(5, 6), use biological translation machinery to produce peptide-based libraries, hits from these libraries often lack key drug-like properties, particularly cell permeability. Though recent advances in library design, such as backbone N-methylation(7), have improved scaffold lipophilicity in mRNA display, codon table constraints still necessitate longer sequences to fully utilize the available space.

DNA-encoded libraries (DELs) offer an alternative approach but face similar challenges in producing drug-like molecules, particularly for intracellular targets, as DELs tend to enrich compounds with poor membrane permeability(8, 9). This selection bias is especially problematic for larger constructs beyond Lipinski's Rules of 5 (bRo5)(10), where fine-tuning lipophilicity is critical. While algorithmic predictions of lipophilicity exist, these 2D atomistic calculations cannot capture conformational effects exhibited by larger molecules like peptide macrocycles(11). Despite over a decade of DEL technology development, no method exists to measure physical properties of encoded molecules within a DNA-encoded library.

DELs are produced through split-pool synthesis using short DNA oligonucleotides to encode the building blocks (BBs) that are incorporated at each chemical step(12, 13). While this approach can draw upon a virtually unlimited set of BBs leading to the construction of billion-member, non-peptidic libraries(14-16), synthetic challenges persist. Truncations are compounded throughout synthesis(17), reducing the representation of properly synthesized constructs. Recent efforts to address this challenge include strategies to improve library purity(18, 19), enable reaction monitoring for macrocycle formation(20), and identify problematic chemistry affecting DNA tag amplification(21). However, despite these improvements(22, 23), no direct method exists for assessing synthetic fidelity on a library-wide basis.

Fast-gradient and multiplexed chromatographic techniques have enabled relatively high-throughput determination of permeability-relevant lipophilicity for small molecules and complex peptides(24-26). However, these techniques are limited to analysis of relatively simple mixtures of mass-unique compounds. Here, we present LC-seq, a new chromatographic DEL selection strategy that simultaneously evaluates synthetic efficiency and permeability-relevant lipophilicity across a proof-of-concept 120,000 member library, enabling empirical determination of drug-like properties on an unprecedented scale.

Method Development and Validation with Model Systems. We first investigated whether DNA-small molecule conjugates could be chromatographically separated based on the physicochemical properties of the pendant small molecules (Fig. 1). A series of alkynes spanning a range of lipophilicities (compounds **1-12**) were conjugated to an azido-functionalized 28-base pair (bp) DNA headpiece via copper-catalyzed click chemistry (Fig. 1A, Fig. S1-S2). Using standard ion-pairing conditions on a reverse-phase C18 column (Table S1), we observed excellent peak shapes and retention times (represented as $\log k'$, the logarithm of the chromatographic capacity factor) which correlated strongly with their calculated lipophilicities (ALogP) ($R^2 = 0.956$) (Fig. 1B,C). Notably, small structural changes produced significant retention time differences – a single methylene difference between compounds **6** and **8** resulted in a 6-minute shift. To validate this approach with more complex structures, we examined 48-bp DNA conjugates of cyclic peptides containing systematic variations in side-chain composition (compounds **13-20**) (Fig. 1 D-F). Retention times again correlated strongly with calculated lipophilicities ($R^2 = 0.925$).

To determine whether more subtle conformational effects due to backbone geometry can lead to retention time differences for cyclic peptide-DNA conjugates, we covalently attached a model 48-bp DNA tag to four cyclic peptide diastereomers whose membrane permeabilities are strongly influenced by the stereochemistry of a single Leu residue (compounds **21-24**) (**Figure 1g**)(27). Retention times of

the cyclic peptide-DNA conjugates reflected the permeability differences within this series, with an 11-minute difference between two epimers that recapitulated the 40-fold difference in their Caco-2 cell permeabilities (**Figure 1h,i**). These results indicate that simple reverse-phase chromatography can readily separate diastereomers with identical calculated lipophilicities, even when attached to a 48-bp duplex DNA barcode.

Macrocyclic Peptide DEL Design. To test whether DNA-conjugated macrocycles can be separated chromatographically in the context of a fully encoded library, we synthesized a 3-cycle DEL of click-cyclized peptides comprised of BBs with a variety of lengths and backbone geometries (Fig. 2A). The DEL was synthesized using 32 BBs in cycles 1 and 2 and 58 BBs in cycle 3, resulting in a 59,392-member library. BBs consisted of α -amino acids, N-methyl amino acids, β -homoamino acids, peptoids (N-alkyl glycines), and backbone-diverse tripeptides. All stereochemical variants for each BB were included to sample a wide range of backbone conformational space, and aliphatic side chains were selected to sample a range of lipophilicities and different degrees of branching. Inclusion of single amino acids and tripeptides at each position resulted in a series of ring sizes (5-, 7-, 9- and 11-mers) that span a molecular weight range between 500 and 1200 Da (Fig. S3).

Every BB at each cycle was encoded with three unique DNA tags, providing 27 redundant encoding sequences per compound. Additionally, in order to track synthetic truncations, at each chemistry cycle three pools received a set of "null" barcodes but were not coupled to a BB at that step. After the three building block cycles, the pooled library was N-terminally modified with azido acetic acid. Half the library underwent copper-mediated click cyclization while half remained uncyclized, and the cyclized and linear pools were encoded separately using unique DNA barcodes. The completed library, including nulls, contained 128,502 compounds. A partially single-stranded tag containing both PCR priming regions and a unique molecular identifier (UMI) was added to enable PCR amplification and removal of duplicate sequences resulting from PCR, respectively (Fig. S4).

Separation and NGS of fractions produces chromatograms based on DNA sequencing counts. For chromatographic analysis, we combined the linear and cyclized sub-libraries in a 1:1 mixture. We injected 50 pmol (3.01×10^{13} molecules) of the mixture onto an analytical RP-HPLC column, separated the mixture into 96 fractions, and quantified the amplifiable DNA in each fraction by qPCR. In independent experiments, we performed the separation using two different elution conditions (Table S1). In both conditions, the library showed broad elution profiles consistent with the hydrophobic nature of the encoded peptides, contrasting sharply with the narrow elution profile of a blank library containing the same permuted DNA sequence but lacking any pendant warheads (Fig. 2B). The less aggressive gradient (10-40% acetonitrile gradient in aqueous TEAA buffer) was selected for sequencing based on its improved separation of late-eluting compounds and more uniform distribution of molecules across the fractions. Fractions were PCR-amplified using unique next-generation sequencing (NGS) adapters to encode each fraction. The encoded fractions were combined and sequenced in a single NextSeq run that produced 1.2×10^9 reads. UMI-based deduplication yielded a total of 5.1×10^8 unique reads across all fractions.

We observed an average of $5.3 (\pm 1.2) \times 10^6$ unique counts per fraction, indicating good representation of the library across the gradient. Significant count enrichment in specific fractions (average sequence count maximum of 248 counts, ~ 10 -fold above baseline) enabled construction of compound-specific LC-seq chromatograms that revealed both the synthetic fidelity and lipophilicity of every library member. For example, two heptamer library members that differed by a single residue, containing either a Nva or Phe at the cycle 1 position, were well separated chromatographically (Fig. 2C). The Phe-containing compound eluted ~ 7 -minutes after the Nva variant, consistent with their relative lipophilicities.

Initial analysis revealed that 48% of the compounds showed identical elution profiles for their linear and cyclized versions, suggesting cyclization failure for these library members (Fig. 2D). Further

investigation revealed that compounds containing either an N-methyl or peptoid residue at the N-terminus failed to couple with azido acetic acid, preventing cyclization (Fig. S5). After filtering these compounds, 30,720 cyclized sequences remained for detailed analysis.

Null-Informed Model to Guide Retention Time Assignment and Profile Reactivity. To rigorously assess synthetic fidelity across the entire library, we developed an algorithm for comparing observed peaks to associated null-encoded truncation products. The inclusion of chemical truncations, encoded with a unique set of null DNA barcodes (*N*) at each synthesis cycle, ensured that all singly and doubly truncated synthetic failures were present in the library and were encoded with the same length of DNA as that of a complete molecule (CM) (Fig. 3A-C). Comparison of the LC-seq trace of an expected CM with those of its corresponding nulls enabled identification of the expected product and its associated synthetic truncations. For example, for the library member cyclo[azidoAc-Leu(NMe)-N-isobutylGly-Pro-Phe-Leu-Prg], consisting of the three building blocks BB55, BB03, and BB01 (Fig. 3B), a major late-eluting peak was observed in the CM chromatogram, which we assigned to the expected heptameric macrocycle (asterisk in Fig. 3B). The CM chromatogram also contained minor peaks corresponding to the two single null truncations indicating loss of either BB03 or BB55. Each of these encoded null truncation chromatograms also contained peaks that aligned with associated double null truncations, providing additional confidence in the identification of these products (Fig. 3B). The LC-seq chromatogram of the closely related CM cyclo[azidoAc-Val-Phe-Leu-Prg] contains a major peak which perfectly aligns with the single null truncation comprised of BB01 and BB03 (Fig. 3C), indicating a failed coupling to BB05 in the third cycle of chemistry.

After filtering out the uncyclized compounds, 10,007 compounds (33%) contained major peaks that matched truncation products, indicating synthesis failure, while 20,643 compounds showed unique, non-null-matching major peaks, suggesting successful synthesis for these library members. The analysis revealed BB-specific trends in the chemistry. For example, peptoid BB11 showed poor coupling as a C-terminal nucleophile with all BBs except the two β -homoamino acids (BB09 and BB10) likely due to reduced steric hindrance of these active esters (Fig. 3D, red circle). The less sterically hindered peptoid, BB12, showed higher coupling efficiency as the N-terminal nucleophile in the second synthetic step (Fig. 3D, green circle).

Library-wide Correlation with Drug-like Properties. The library was designed so that each scaffold contains multiple R-group variants, leading to a "liposcan" for each scaffold. There was a strong correlation between LC-seq retention times and calculated lipophilicity (AlogP) within a given scaffold, as well as significant differences in retention time among different scaffolds at the same ALogP, consistent with the results from the model compounds (Fig. 1G-I) and providing further validation that the library was decoded correctly. For example, two scaffolds that vary in stereochemistry at a single residue (D-Leu vs. L-Leu) showed significantly different retention times among matched pairs with the same combination of R groups; however, within the same scaffold the retention times were highly correlated with the ALogP values of the individual liposcan variants (Fig. 4A,B).

To further confirm the sequencing accuracy of the filtered LC-seq chromatograms and test whether LC-seq retention times reflect membrane permeability in more biologically relevant assays, we individually resynthesized 24 compounds spanning different ring sizes and cleaved them from the solid support as their C-terminal ethyl esters(28) (Fig. 4C). The on-DNA retention times showed excellent correlation with off-DNA measurements across all ring sizes ($R^2 = 0.936$), confirming the LC-seq-assigned structural identities of these compounds (Fig. 4D).

We then evaluated the passive membrane permeabilities of the 24 resynthesized compounds using the parallel artificial membrane permeability assay (PAMPA). While calculated AlogP values showed poor correlation with permeability (Fig. 5A), particularly for stereoisomers, on-DNA retention times showed strong, ring size-dependent correlations with PAMPA permeability (Fig. 5B). The permeability gain per unit increase in lipophilicity remained consistent across 5-mers, 7-mers, and 9-mers, with the 11-mers

showing reduced gains, likely approaching size limitations for passive membrane permeability. As expected, the lipophilicity threshold required for high permeability increased with ring size, consistent with known MW-permeability relationships(29). The PAMPA measurements showed good agreement with MDCK cell permeabilities, supporting broader relevance for cellular penetration (Fig. 5C). These results demonstrate that the LC-seq platform provides not only quality control data but also biologically relevant insights into compound properties, enabling more informed DEL optimization.

Implications for extension to other drug-like properties. We introduce a transformative chromatographic strategy that enables direct assessment of both synthetic quality and drug-like properties for individual members of DNA-encoded libraries. This approach addresses two critical challenges in DEL-based drug discovery: identifying false positives from failed synthesis and predicting membrane permeability prior to resource-intensive resynthesis. The method's ability to track reaction efficiency for all building block combinations provides unprecedented insight for library design, while empirical measurements of lipophilicity enable rational prioritization of compounds for development. Although demonstrated here with a focused 120,000-member library, our results suggest that the approach is scalable to million-member libraries with current sequencing capabilities and would be applicable to DNA-encoded libraries of small molecules in addition to peptidic macrocycles. We are investigating whether LC-seq can be applied to the analysis of mRNA display libraries, and more generally, to encoded libraries containing more polar and chemically complex side chains.

The LC-seq platform could be adapted to evaluate additional drug-like properties through strategic modifications of the separation conditions. For metabolic stability assessment, the library could be exposed to liver microsomes or other specific metabolizing enzymes (e.g., proteases) before chromatographic analysis, with changes in retention time and peak profiles indicating metabolic modifications. Sequential time-point analysis would enable quantitative stability measurements across the library. The large-scale property measurements enabled by LC-seq could dramatically enhance AI models' predictive capabilities, allowing more accurate evaluation of the vast virtual chemical space encompassed by larger macrocycles. This technology has immediate applications in library design and hit triage, potentially transforming the efficiency of DEL-based drug discovery.

References and Notes

1. Y. Fan *et al.*, Encoding and display technologies for combinatorial libraries in drug discovery: The coming of age from biology to therapy. *Acta. Pharm. Sin. B* **14**, 3362-3384 (2024).
2. A. Gironda-Martinez, E. J. Donckele, F. Samain, D. Neri, DNA-Encoded Chemical Libraries: A Comprehensive Review with Successful Stories and Future Challenges. *ACS Pharmacol. Transl. Sci.* **4**, 1265-1279 (2021).
3. Y. Huang, M. M. Wiedmann, H. Suga, RNA Display Methods for the Discovery of Bioactive Macrocycles. *Chem. Rev.* **119**, 10360-10391 (2018).
4. S. E. Iskandar, A. A. Bowers, mRNA Display Reaches for the Clinic with New PCSK9 Inhibitor. *ACS Med. Chem. Lett.* **13**, 1379-1383 (2022).
5. W. Jaroszewicz, J. Morcinek-Orlowska, K. Pierzynowska, L. Gaffke, G. Wegrzyn, Phage display and other peptide display technologies. *FEMS Microbiol. Rev.* **46**, 1-25 (2022).
6. J. T. Hampton, W. R. Liu, Diversification of Phage-Displayed Peptide Libraries with Noncanonical Amino Acid Mutagenesis and Chemical Modification. *Chem. Rev.* **124**, 6051-6077 (2024).
7. M. Tanada *et al.*, Development of Orally Bioavailable Peptides Targeting an Intracellular Protein: From a Hit to a Clinical KRAS Inhibitor. *J. Am. Chem. Soc.* **145**, 16610-16620 (2023).
8. A. P. Silvestri *et al.*, DNA-Encoded Macrocyclic Peptide Libraries Enable the Discovery of a Neutral MDM2-p53 Inhibitor. *ACS Med. Chem. Lett.* **14**, 820-826 (2023).

9. T. Passioura, W. Liu, D. Dunkelmann, T. Higuchi, H. Suga, Display Selection of Exotic Macrocyclic Peptides Expressed under a Radically Reprogrammed 23 Amino Acid Genetic Code. *J. Am. Chem. Soc.* **140**, 11551-11555 (2018).
10. C. A. Lipinski, F. Lombardo, B. W. Dominy, P. J. Feeney, Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv. Drug Deliv. Rev.* **23**, 3-25 (1997).
11. M. R. Naylor *et al.*, Lipophilic Permeability Efficiency Reconciles the Opposing Roles of Lipophilicity in Membrane Permeability and Aqueous Solubility. *J. Med. Chem.* **61**, 11169-11182 (2018).
12. M. A. Clark *et al.*, Design, synthesis and selection of DNA-encoded small-molecule libraries. *Nat. Chem. Biol.* **5**, 647-654 (2009).
13. S. Brenner, R. A. Lerner, Encoded Combinatorial Chemistry. *Proc. Natl. Acad. Sci. USA* **89**, 5381-5383 (1992).
14. J. P. Phelan *et al.*, Open-Air Alkylation Reactions in Photoredox-Catalyzed DNA-Encoded Library Synthesis. *J. Am. Chem. Soc.* **141**, 3723-3732 (2019).
15. Y. Ruff *et al.*, An Amphiphilic Polymer-Supported Strategy Enables Chemical Transformations under Anhydrous Conditions for DNA-Encoded Library Synthesis. *ACS Comb. Sci.* **22**, 120-128 (2020).
16. V. K. Siripuram, Y. K. Sunkari, T. L. Nguyen, M. Flajolet, DNA-Compatible Suzuki-Miyaura Cross-Coupling Reaction of Aryl Iodides With (Hetero)Aryl Boronic Acids for DNA-Encoded Libraries. *Front. Chem.* **10**, 1-7 (2022).
17. A. L. Satz, R. Hochstrasser, A. C. Petersen, Analysis of Current DNA Encoded Library Screening Data Indicates Higher False Negative Rates for Numerically Larger Libraries. *ACS Comb. Sci.* **19**, 234-238 (2017).
18. M. P. Keller, Dimitar; Gloger, Andreas; Dietschi, Bastien; Jobin, Kilian; Gradinger, Timon; Martinelli, Adriano; Plais, Louise; Onda, Yuichi; Neri, Dario; Scheuermann, Jörg, Highly pure DNA-encoded chemical libraries by dual-linker solid-phase synthesis. *Science* **384**, 1259-1265 (2024).
19. R. M. Franzini *et al.*, "Cap-and-Catch" Purification for Enhancing the Quality of Libraries of DNA Conjugates. *ACS Comb. Sci.* **17**, 393-398 (2015).
20. A. Roy, E. Koesema, T. Kodadek, High-Throughput Quality Control Assay for the Solid-Phase Synthesis of DNA-Encoded Libraries of Macrocycles. *Angew. Chem. Int. Ed. Engl.* **60**, 11983-11990 (2021).
21. A. S. Ratnayake *et al.*, A Solution Phase Platform to Characterize Chemical Reaction Compatibility with DNA-Encoded Chemical Library Synthesis. *ACS Comb. Sci.* **21**, 650-655 (2019).
22. O. B. C. Monty, N. Simmons, S. Chamauri, M. M. Matzuk, D. W. Young, Solution-Phase Fmoc-Based Peptide Synthesis for DNA-Encoded Chemical Libraries: Reaction Conditions, Protecting Group Strategies, and Pitfalls. *ACS Comb. Sci.* **22**, 833-843 (2020).
23. R. Rama-Garda *et al.*, Normalization of DNA encoded library affinity selection results driven by high throughput sequencing and HPLC purification. *Bioorg. Med. Chem.* **40**, 116178 (2021).
24. G. Koch *et al.*, Chromatographic Determination of Permeability-Relevant Lipophilicity Facilitates Rapid Analysis of Macrocyclic Peptide Scaffolds. *J. Med. Chem.* **67**, 19612-19622 (2024).
25. A. Watanabe, S. Uehara, T. Akazawa, M. Fujiu, Optimization of Cyclic Peptide Property Using Chromatographic Capacity Factor on Permeability of Passive Cell Membrane and Human Induced Pluripotent Stem Cell-Derived Intestinal Membrane. *J. Pharm. Sci.* **111**, 1879-1886 (2022).
26. G. Caron *et al.*, A Fast Chromatographic Method for Estimating Lipophilicity and Ionization in Nonpolar Membrane-Like Environment. *Mol. Pharm.* **13**, 1100-1110 (2016).

27. W. M. Hewitt *et al.*, Cell-Permeable Cyclic Peptides From Synthetic Libraries Inspired by Natural Products. *J. Am. Chem. Soc.* **137**, 715-721 (2015).
28. R. Turner, R. Weber, R. S. Lokey, Direct Conversion of Resin-Bound Peptides to C-Terminal Esters. *Org. Lett.* **12**, 1852-1855 (2010).
29. C. R. Pye *et al.*, Nonclassical Size Dependence of Permeation Defines Bounds for Passive Adsorption of Large Drug Molecules. *J. Med. Chem.* **60**, 1665-1672 (2017).

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Conceptualization: GK, RSL, MFL, NT

Methodology: GK, MFL, RSL

Investigation: GK, MFL

Chemistry: GK, MS, SBA, CC, PZ, JT, LG

Molecular Biology: MFL, KCL, GK

Permeability Assays: AE, GK

Computations: AM, NC, SL, GK

Project administration: GK, RSL, NT

Supervision: RSL, NT, LG, JT

Writing: GK, RSL

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Supplementary Materials

Supplementary Text

Figs. S1 to S5

Table S1

Materials and Methods

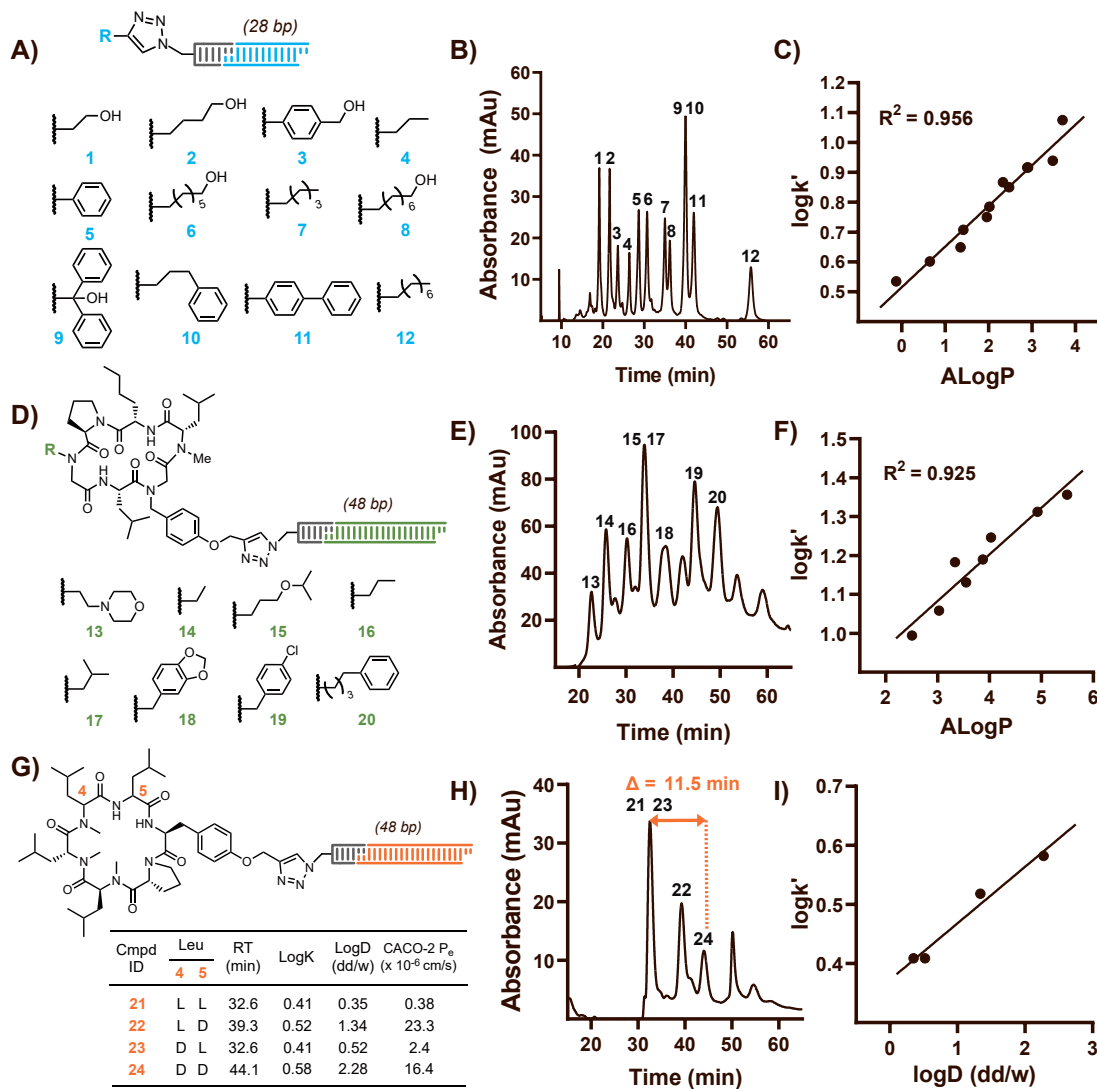


Figure 1: Chromatographic behavior of model DNA-conjugates demonstrates correlation between retention time and molecular properties. (A) Structures of alkyne-modified small molecules (1-12) conjugated to 28-bp DNA via copper-catalyzed click chemistry. (B) HPLC chromatogram showing separation of a coinjection of DNA-conjugates 1-12 under standard ion-pairing conditions. (C) Correlation between retention times ($\log k'$) and calculated ALogP values ($R^2 = 0.956$). (D) Structure of cyclic peptide scaffold with varying N-alkyl-Gly substituents (13-20) conjugated to 48-bp DNA. (E) HPLC separation of a coinjection of peptide-DNA conjugates 13-20. (F) Correlation between retention times ($\log k'$) and ALogP values ($R^2 = 0.925$). (G) Structure and stereochemical variants of a model cyclic peptide series (21-24) with known membrane permeability differences. (H) HPLC separation showing 11-minute retention time difference between epimers. (I) Correlation between chromatographic lipophilicity ($\log k'$) and experimental LogD values for 21-24.

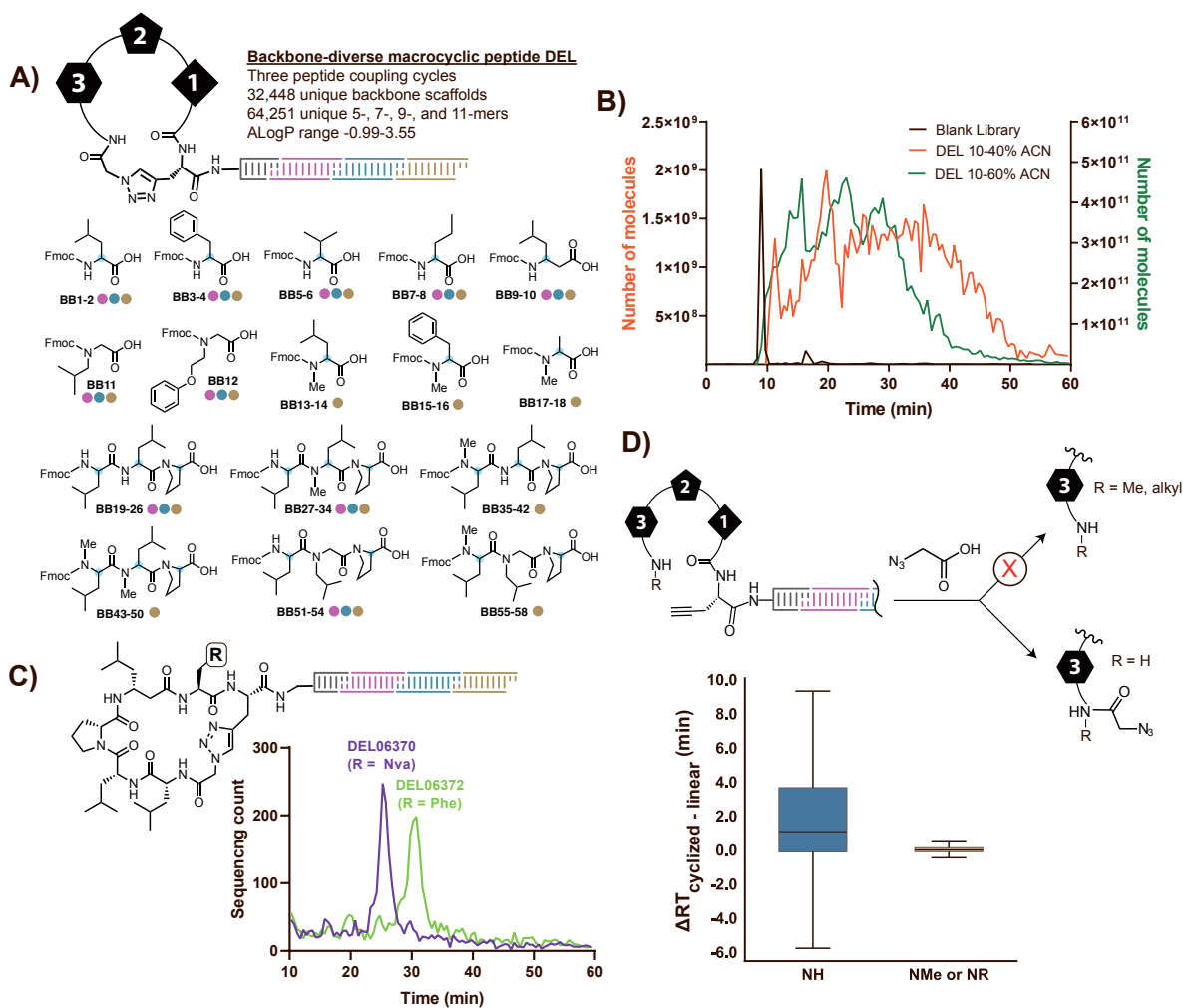


Figure 2: DNA-encoded library design and chromatographic analysis workflow. (A) Library architecture showing building block diversity and incorporation positions. Colored circles indicate building block usage in different cycles. (B) qPCR quantification of library fractions collected during chromatographic separation under different acetonitrile gradients (10-40% orange, 10-60% green) compared to DNA-only blank library (black). (C) Example chromatographic profiles for two library members differing by a single residue (Nva vs. Phe), demonstrating sequence-specific retention behavior. (D) Scheme and retention time data demonstrating disparate cyclization efficiency between DEL members with either a primary or secondary N-terminal amine.

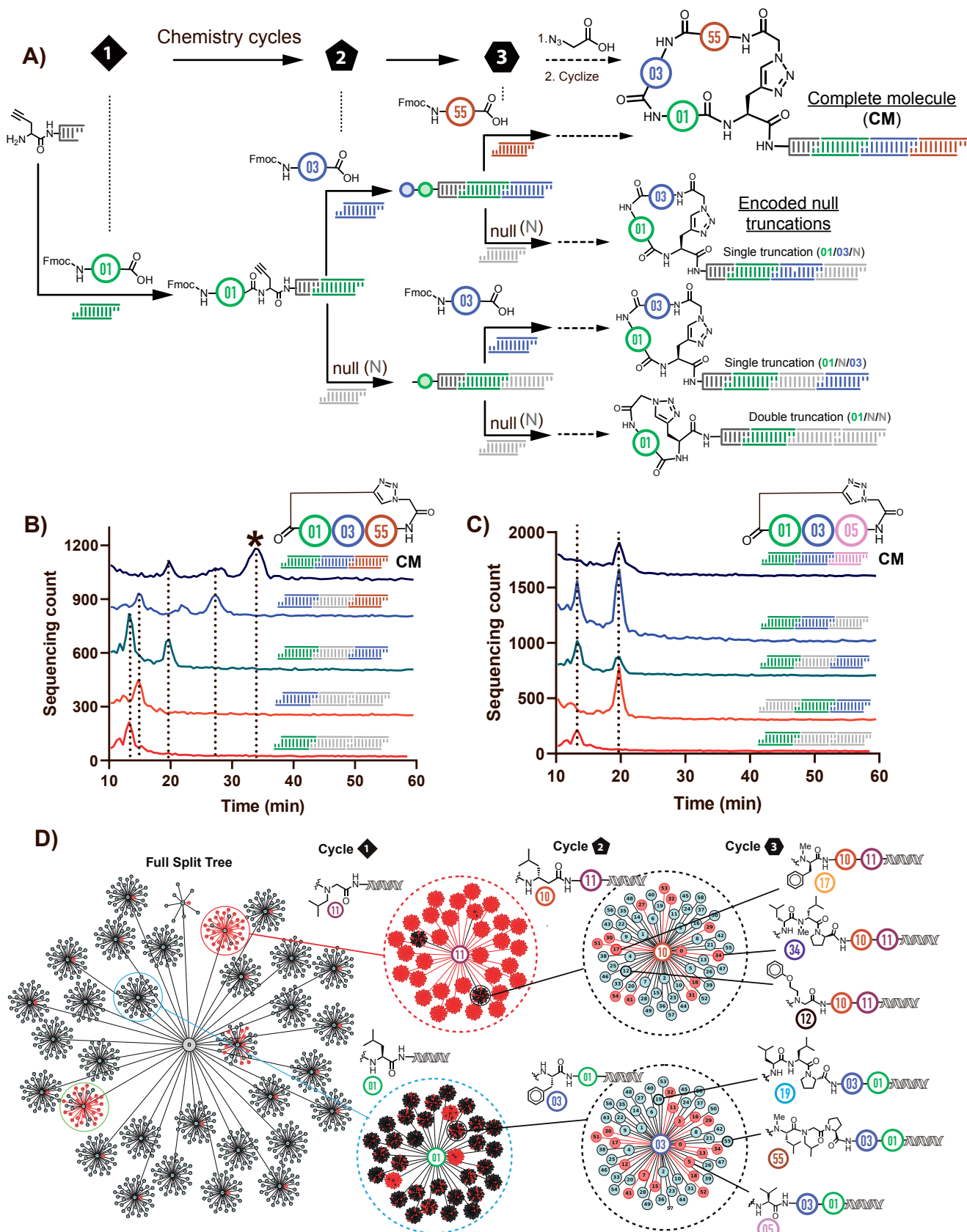


Figure 3: Null-informed analysis enables library-wide assessment of synthetic fidelity. (A) Schematic showing incorporation of null encoding tags and resulting truncation products. (B) Example chromatogram for successful synthesis showing product peak (*) distinct from truncation products. (C) Example chromatogram for failed synthesis showing product peak matching truncation products. (D) Split-tree visualization of library-wide synthetic efficiency where node color indicates synthesis success (blue) or failure (red). Highlighted clusters (red, blue, green circles) demonstrate building block-specific trends in reaction efficiency.

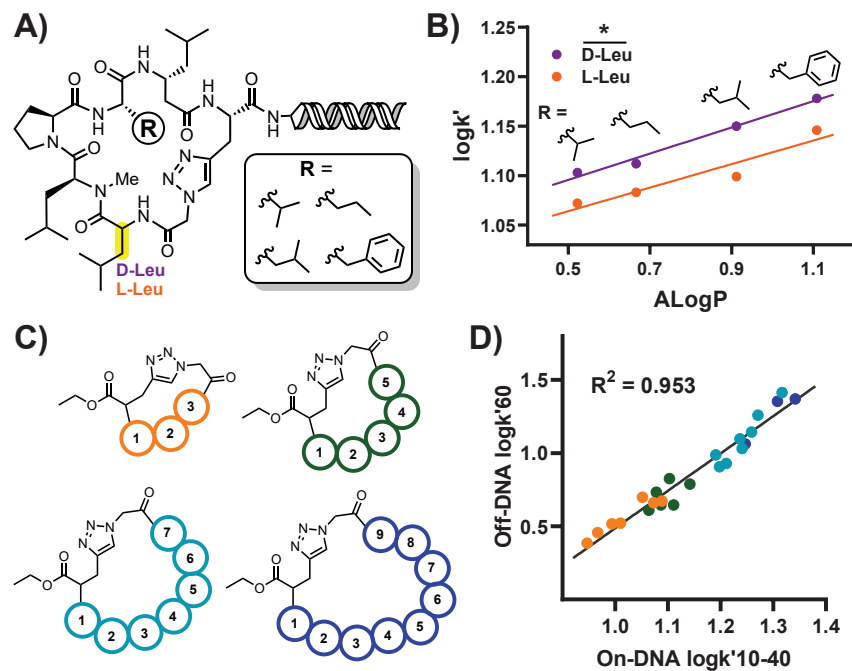


Figure 4: Assessment of on-DNA lipophilicity measurements from the DEL. (A) Structures of two scaffolds from the DEL which differ by inversion of a single stereocenter (Leu, highlighted yellow) with each scaffold containing four unique liposcan members. (B) Relationship between on-DNA retention times ($\log k'$) and calculated lipophilicity (ALogP). Each scaffold displays an excellent correlation with a significant difference in lipophilicity between each scaffold for all four members. (C) Simplified structures of the resynthesized compound sets. The connection to the DNA has been replaced with an ethyl ester for off-DNA testing. (D) Relationship between on-DNA capacity factors and off-DNA capacity factors. An excellent correlation ($R^2 = 0.936$) is observed across all ring sizes.

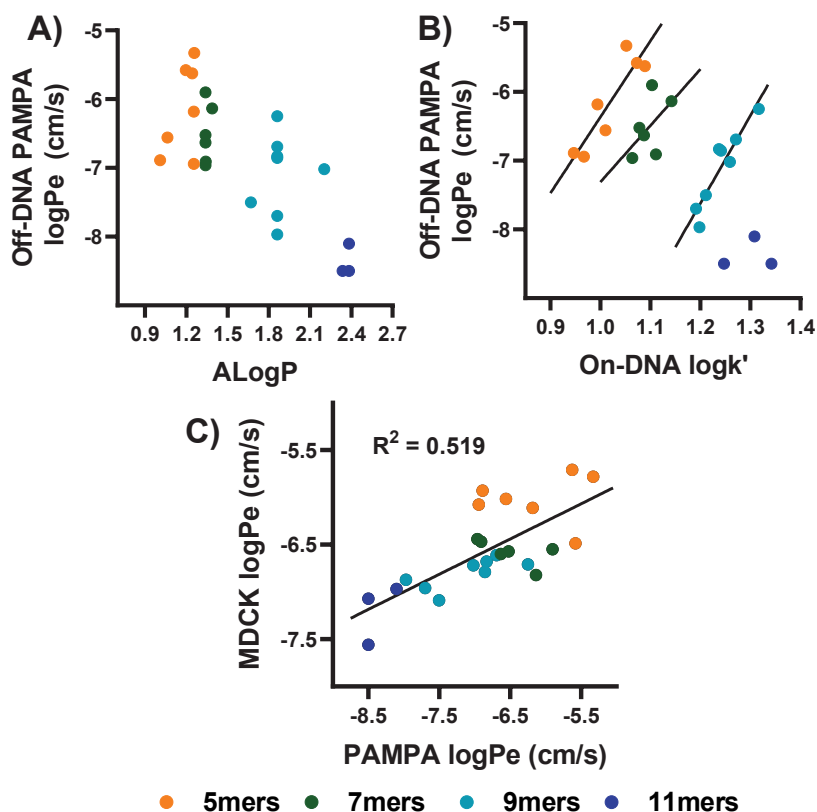


Figure 5: Relationship between on-DNA lipophilicity and passive permeability in PAMPA and MDCK for the 24 resynthesis compounds. Data are colored by ring size. A) Relationship between PAMPA permeability and AlogP. Vertical lines indicate isomers or stereoisomers that have identical calculated lipophilicities. B) Relationship between off-DNA PAMPA permeability and on-DNA capacity factors (retention time). Linear regressions are shown (black lines) for each ring size except the undeca peptides. C) Relationship between MDCK and PAMPA permeability. Linear regression (black line) and coefficient of determination are shown across all ring sizes.

